



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

ASSIGNMENT No: 2

TITLE: Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.



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Exercise

1. Create a Student class using default and parameterized constructors to initialize student name and roll number.

Code

```
experiments - experiment1/src/experiment1/Experiment1.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer
  experiment1
    > JRE System Library [JavaSE-21]
    > src
      experiment1
        > Experiment1.java
        > module-info.java

Experiment1.java
1 package experiment1;
2
3 class Experiment1 {
4     String name;
5     int rollNo;
6
7     // Default Constructor
8     Experiment1() {
9         name = "Unknown";
10        rollNo = 0;
11    }
12
13    // Parameterized Constructor
14    Experiment1(String name, int rollNo) {
15        this.name = name;
16        this.rollNo = rollNo;
17    }
18
19    // Copy Constructor
20    Experiment1(Experiment1 s) {
21        this.name = s.name;
22        this.rollNo = s.rollNo;
23    }
24
25    // Display Method
26    void display() {
27        System.out.println("Student Name: " + name);
28        System.out.println("Roll No: " + rollNo);
29    }
30
31    public static void main(String[] args) {
32
33    }
34    }
```

```
31    public static void main(String[] args) {
32
33        // Default Constructor
34        Experiment1 s1 = new Experiment1();
35        System.out.println("Default Constructor:");
36        s1.display();
37
38        // Parameterized Constructor
39        Experiment1 s2 = new Experiment1("Karan Dave", 507);
40        System.out.println("\nParameterized Constructor:");
41        s2.display();
42
43        // Copy Constructor
44        Experiment1 s3 = new Experiment1(s2);
45        System.out.println("\nCopy Constructor:");
46        s3.display();
47    }
48    }
```



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Output

```
Default Constructor:  
Student Name: Unknown  
Roll No: 0  
  
Parameterized Constructor:  
Student Name: Karan Dave  
Roll No: 507  
  
Copy Constructor:  
Student Name: Karan Dave  
Roll No: 507
```



2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

Code

```
1 class Mobile
2 {
3     String model;
4     double price;
5     // Default Constructor
6     Mobile()
7     {
8         model = "Huawei Pura 80";
9         price = 6499;
10    }
11    // Parameterized Constructor
12    Mobile(String m, double p)
13    {
14        model = m;
15        price = p;
16    }
17    // Copy Constructor
18    Mobile(Mobile obj)
19    {
20        model = obj.model;
21        price = obj.price;
22    }
23    void display()
24    {
25        System.out.println("Model : " + model);
26        System.out.println("Price : ₹" + price);
27        System.out.println();
28    }
29 }
```

```
30 public class MobileInventory
31 {
32     public static void main(String[] args)
33     {
34         // Using Default Constructor
35         Mobile m1 = new Mobile();
36
37         // Using Parameterized Constructor
38         Mobile m2 = new Mobile("Meizu Note 16", 1599);
39
40         // Using Copy Constructor
41         Mobile m3 = new Mobile(m2);
42
43
44         System.out.println("Mobile 1 Details");
45         m1.display();
46
47         System.out.println("Mobile 2 Details");
48         m2.display();
49
50         System.out.println("Duplicate Mobile 3 Details");
51         m3.display();
52     }
53 }
```



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Output

Mobile 1 Details

Model : Huawei Pura 80

Price : ₹6499.0

Mobile 2 Details

Model : Meizu Note 16

Price : ₹1599.0

Duplicate Mobile 3 Details

Model : Meizu Note 16

Price : ₹1599.0